



The role of perceived risk and information security on customers' acceptance of service robots in the hotel industry

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ABSTRACT

This study proposed and tested a theoretical framework that investigated the influences of perceived risk and information security on hotel customers' intention to use service robots. In addition, the impacts of self-efficacy, innovativeness, and facilitating conditions on perceived risk and information security were examined. Structural equation modeling (SEM) was used to test the proposed model by utilizing data collected from eleven countries including the United States, United Kingdom, Turkey, Spain, Romania, Japan, Israel, India, Greece, Canada, and Brazil. The study results demonstrated that perceived risk had a negative impact on customers' intention to use service robots while information security had a positive impact. In addition, the study results indicated that self-efficacy negatively influenced perceived risk, and positively influenced perceived information security; and innovativeness and facilitating condition positively influenced information security. The study findings offer several important contributions to the hospitality robotics technology adoption literature and present valuable implications for hospitality practitioners and service robot vendors.

1. Introduction

The progress in robotic technologies is setting the stage for unprecedented levels of robot integration and adoption both at industrial and consumer levels. The number of patent filings for robotic artificial intelligence (AI), for instance, reached a record-high annual growth rate of 55% between 2013 and 2016 (WIPO, 2019). The global service-robot market is projected to achieve a market size of \$110.4 billion by 2026, representing a remarkable growth from its 2022 value of \$45.5 billion (Strategy, 2022). Meanwhile, the hospitality service robot market stood at \$295.5 million in 2020, and forecasts suggest that it will reach \$3.08 billion by 2030, with a notable compound annual growth rate (CAGR) of 25.5% from 2021 to 2030 (Anil and Roshan, 2021). In June 2021, Hyundai announced that the company completed its acquisition of

Boston Dynamics, an engineering and robotics design company (Heater, 2021). The company paid \$880 million for an 80% stake in the company, a quadruple increase in value since Boston Dynamics's acquisition by SoftBank, a Japanese investment firm, from Google only three years ago (Bartneck, 2022). As the data and developments suggest, we are at an inflection point in human history where productivity gains and services rendered are increasingly reliant on robotics and AI-based technologies (Ivanov et al., 2019; Tung and Law, 2017; Wirtz et al., 2018).

The term "t" was first introduced by Czech playwright Karel Čapek in 1920, in his work titled "Rossum's Universal Robots" (Čapek, 2004). In Čapek's book, the term *ta* denoted forced labor or slave, which later evolved to describe machines that are designed to perform highly repetitive tasks (Lanfranco et al., 2004). Similarly, a service robot can be defined as a "system-based autonomous and adaptable interfaces that

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interact, communicate and deliver service to an organization's customers" (Wirtz et al., 2018, p. 909). Equipped with technologies such as processors, sensors, image processing, biometrics, IoT, service robots are physical entities that are employed in various service settings as front-line agents (Ivanov and Webster, 2020; Ivanov et al., 2022; Paluch et al., 2022).

In the post-Covid-19 era, service robots have proven to be a topic of scholarly interest, particularly in the human-t-interaction (HRI) context (Zeng et al., 2020). Their capacity to reduce the spread of contagious diseases by minimizing and, sometimes, substituting human-to-human interaction has been well recognized (Wan et al., 2020; Kabadayi et al., 2020). The pandemic has caused a shift in the way businesses operate, with many companies looking for ways to reduce human-to-human contact and minimize the risk of spreading the virus (Thomas et al., 2021). This gave rise to robots' adoption in various settings, such as in hospitals (Javaid et al., 2020), airports (Wirtz et al., 2021), and hotels (Jiang and Wen, 2020).

The adoption and use of service robots have been previously studied in different settings including restaurants (Liu et al., 2022), hotels (Shin and Jeong, 2020), hospitals (Park et al., 2021), senior living facilities (Takanokura et al., 2021), and cruise ships (Gonzales et al., 2021). These studies have investigated various factors that influenced customers' behavioral intentions toward service robots such as perceived enjoyment (Go et al., 2020), perceived usefulness (Zhong et al., 2021), perceived ease of use (Seo and Lee, 2021), perceived sociability (Forgas-Coll et al., 2022), perceived trust (Seo and Lee, 2021), hedonic and utilitarian values (Hwang, Choe et al., 2021) (Hu, 2021), performance and effort expectancy (Lin et al., 2020); Song, Zhang, Hu, & Cao, 2022), and utility (Cha, 2020). Yet, there is a lack of research on customers' perceptions of various risk dimensions and information security associated with the use of service robots in the hotel industry. This is a significant gap in the literature given that being served by robots may cause discomfort and anxiety for some hotel customers as they may be concerned about the potential risks involved, such as the physical risk. Services by robots may also lead to concerns of financial risk as their perceived financial cost may not align with their perceived value. Performance risk, meanwhile, may arise from the customers' anticipation of service failure when they are served by the robots. Perceptions of psychological risk, meanwhile, may arise as a consequence of customers' psychological discomfort or anxiety resulting from their interaction with service robots.

Theories on risk perception, e.g., Bauer (1960), Pollatsek and Tversky (1970), Kaplan and Garrick (1981), and Rescher (1983) suggest that human actions entail a certain level of uncertainty regarding the effects and outcomes of these actions. Given that robots can now serve as a substitute for many human activities and engage in various interactions with humans, investigating the risks customers associated with robots becomes inevitable (Yagoda and Gillan, 2012). Customers' risk perceptions have been previously studied in the robotics context, e.g., Cha (2020), Chi et al. (2020), and El-Said and Al Hajri (2022). However, limited research exists on various risk dimensions, such as physical risk, financial risk, performance risk, and psychological risk, in the service robot context.

Information security, on the other hand, is related to customers' perceptions of transaction security when personal and payment information is provided to the service robots by the customers (Chang and Chen, 2009). Similar to the exchange of personal information with self-service technologies (SST), customers are now required to provide their personal information to service robots in various service areas, such as the front desk, food & beverage outlets, room service, and concierge. Therefore, providing a sense of information security and confidentiality to customers is vital for robotic technology investments to reach their full potential.

Choi and Wan (2021) posited that service robot technologies reached the level of intelligence that is necessary to provide more intuitive and spontaneous human-t interactions (HRIs). In line with this proposition,

the present study explores the risk and information security perceptions of hotel customers by evaluating the factors, such as self-efficacy, innovativeness, and facilitating conditions, that influence these perceptions. As a personal trait, self-efficacy has been previously proposed as a mitigator of customers' risk perceptions of products or services (Kim and Kim, 2005). Further, a positive relationship can also be proposed between the customers' level of self-efficacy and their perceptions of information security. According to Dabholkar and Bagozzi (2002), customers with higher levels of self-efficacy tend to show higher levels of intention to use technology-based services. This is of particular interest to hospitality organizations since customers who are comfortable when interacting with service robots and have higher levels of self-efficacy are more likely to use service robots.

Innovativeness is another personal trait that is pivotal to customers' intentions of using advanced technologies, such as service robots. It is the defining characteristic of early adopters when a new product or technology is introduced (Rogers, 1995). Innovativeness has been defined as a customer's proclivity to risk-taking by the previous literature (Truong, 2013), which is hypothesized to have a negative impact on customers' perceived risk by the present study. Given the innovative customers' level of comfort with new technologies, they are also expected to be less concerned about information security during their interactions with service robots. Facilitating condition, meanwhile, can be defined as the technological infrastructure the organizations provide to their customers to facilitate the use of their systems (Venkatesh et al., 2003). Existing literature provides sufficient evidence for facilitating condition's role in mitigating customers' risk perceptions while elevating their perceptions of information security with new technologies (Lu et al. (2019).

Based on the discussion provided above, the present study investigates the factors that affect customers' intention to use service robots in the hotel industry. More specifically, this study investigated the roles of perceived risk and information security on customers' intentions to use service robots in hotels. Furthermore, the impacts of self-efficacy, innovativeness, and facilitating conditions on perceived risk and information security were evaluated. Compared with previous literature on service robots, this research is one of the first cohorts that proposed and validated the four sub-dimensions of perceived risk in the robotic service context. Additionally, the present study provided empirical evidence to support the direct impacts of perceived risk and information security on hotel customers' intentions to use service robots. Furthermore, this study is among the first to examine customers' personal traits in the service robot context while investigating technology infrastructure's impact on their risk and information security perceptions.

The conceptual framework developed by the present study has important implications for hospitality practitioners and service t vendors alike. The results generated present several noteworthy conclusions and highlight some key areas where additional research can be conducted before service robots become a global phenomenon in the hospitality industry.

2. Literature review

2.1. Intention to use

Humans are rational and goal-driven processors of information (Heider, 1958). The information that is available to them regarding their circumstances impacts their beliefs, attitudes, intentions, and behaviors (Fishbein & Ajzen, 1975). Although intrinsically related, a distinction should be made between beliefs, attitudes, intentions, and behaviors. As a mental state of intentionality, a belief can be defined as a person's conceptions, ideologies, and perceptions of the world, which guide one's thinking while shaping his or her actions. Attitude refers to the perceptions of an outcome, i.e., favorable vs. unfavorable. Both beliefs and attitudes are a product of past experiences. Intention, however, indicates a person's readiness to perform a particular behavior (Passos et al.,

2013).

In the technology acceptance context, intention refers to the subjective probability that a person will perform a certain behavior (Liu and Huang, 2015). In diffusion and adoption of technology research, customers' use intentions were studied in various contexts, such as social networks (Shin and Kim, 2008), interactive mobile technologies (Wendy Zhu and Morosan, 2014), and NFC-based mobile payment technologies (Ozturk et al., 2017). Intention to use, as an outcome variable, was also examined in robotics contexts in different settings including healthcare (Turja et al., 2020; Vichitkraivin and Naenna, 2021), tourism (Ivanov and Webster, 2019; Tusyadiah, 2020), restaurants (Kim et al., 2023; Sung and Jeon, 2020); (Cha, 2020; Kwak et al., 2021), and hotels (Lee and Kim, 2022; Song et al., 2022). In the present study, intention to use was defined as the degree to which hotel customers intend to use service robots.

2.2. Perceived risk

In marketing research, the theory of perceived risk was originally introduced by Bauer (1960) to investigate its potential in explaining a diverse set of phenomena, such as the search for information and brand loyalty. According to Bauer (1960), consumers engage in purchasing behaviors that have the potential to generate unpleasant consequences, which cannot be foreseen by the consumers. When faced with a purchasing decision that involves risk, the theory of perceived risk posits, consumers choose to mitigate this risk by adopting one of the several options available to them- choosing to become more loyal to a brand is the most conventional behavior (Bauer, 1960). In the consumer behavior context, perceived risk was defined as the amalgam of chance and danger dimensions, where chance meant probability and danger referred to the magnitude of the negative consequence based on the decision (Kogan and Wallach, 1964).

Prior research demonstrated that perceived risk is a multidimensional concept (Kaplan et al., 1974; Park and Tusyadiah, 2017) that arises from various potential losses associated with consumption behavior (Jacoby and Kaplan, 1972; Kaplan and Garrick, 1981). The existing literature identified several dimensions of perceived risk including performance risk, social risk, time risk, financial risk, physical risk, and psychological risk (Littler and Melanthiou, 2006; Stuck et al., 2021). Performance risk can be described as a product failure when the expected performance cannot be delivered as a result of a malfunctioning or breakdown, therefore failing to fulfill the desired goals (Featherman and Pavlou, 2003; Jacoby and Kaplan, 1972). Social risk can be defined as the potential loss of esteem, respect, and/or social status involved in a purchasing or consumption decision (Murray and Schlacter, 1990). Time risk stems from the customers' perceptions of the potential time loss resulting from the use of a product or service (Roselius, 1971). Financial risk can be described as the likely expense associated with the purchase of a product and the ensuing cost of upkeep (Grewal et al., 1994). In simpler terms, it is the notion that the user can lose money due to the interaction or circumstance at hand and incur a financial loss (Stuck et al., 2021). Physical risk is related to the potential physical harm that is associated with using a product or service (Dholakia, 2001). Psychological risk can be defined as the uncertainty that is related to the anticipated emotions following a purchase experience or utilization of a product. This uncertainty may manifest itself as negative emotions, e.g., disappointment, distress and remorse (Hirunyawipada and Paswan, 2006).

A careful consideration of perceived risk could provide valuable insights into assessing customers' use intentions by gauging their perceptions of potential negative outcomes. Given the rise in the employment of service robots in the hospitality industry, recognizing customers' risk perceptions is a key factor for organizations to generate a meaningful return on their technology investments. This study investigated four risk dimensions (i.e., physical risk, financial risk, performance risk, and psychological risk) that are associated with the use of

service robots in hotels.

The physical risk is often related to the consumer's lack of knowledge regarding a new electronic product or technology. This state of knowledge scarcity urges consumers to search for information on the new product or technology as its use may cause physical harm (Dholakia, 2001; Hirunyawipada and Paswan, 2006). Similarly, a service provided by a robot may lead to uncomfortable physical circumstances or concern for potential physical risks. Some customers may also feel physically unsafe when they are served by robots. Pointing to the physical presence of robots, Stuck et al. (2021), for example, emphasized the physical injuries and fatal accidents that have taken place because of robots' actions.

Financial risk is related to the user's concern over potential financial losses, which may lead to the decision to reject a product or technology. Hirunyawipada and Paswan (2006) posited that perceived financial risk is negatively associated with the purchase of consumer electronics. Lee (2009) reached a similar conclusion in a study that explored customers' behavioral intentions to use online banking. The same can be inferred from hotel customers' perceptions of financial risks when they are served by robots. Hotel customers may seek the assurance that they will be getting their money's worth when they are staying at a hotel that employs service robots. Hwang, Kim et al. (2021) finessed this proposition in the robotic restaurant context and concluded that customers financial risk perceptions negatively influenced their attitudes toward using robotic restaurants.

Littler and Melanthiou (2006) approached performance risk from the users' existing cognitive abilities of using the product and knowledge about the technology. This perspective is also relevant in the robotic context as hotel customers may approach service robots with the apprehension that the service provided by robots may not meet their expectations or that something is bound to go wrong. In other words, customers may associate using service robots with risk, which potentially diminishes their willingness to engage with them. Accordingly, evaluating the customers' perceptions accurately and taking a proactive approach is critical for hospitality organizations to successfully implement robots and other artificially intelligent agents in a hotel environment.

According to Littler and Melanthiou (2006), customers' perceptions of self may be negatively affected in the event of a failure of a new technology. In the hotel setting, however, the psychological risk of being served by a robot may start even before the interaction takes place. The notion of being served by a robot may give rise to feelings of discomfort and anxiety for some customers (Stuck et al., 2021). Given the hedonic nature of hospitality experiences (Pine and Gilmore, 1999; Pizam, 2010), any undue psychological burden on customers could adversely affect their intentions to use service robots. Furthermore, misvaluations of customers' risk perceptions and how they induce avoidance behavior (Conchar et al., 2004) is an undesirable scenario for hospitality operators. Accordingly, the present study measured hotel customers' risk perceptions as a second-order construct with four risk dimensions, i.e., physical, financial, performance, and psychological. Thus, the following hypothesis is proposed:

H_1 : Hotel customers' perceived risk toward service robots negatively influences their intention to use service robots.

2.3. Information security

Information security, a component of transaction security, refers to the customers' perceptions of security, which they ascribe to the electronic medium to which they provide their personal information, and how they believe this information is stored and protected (Chang and Chen, 2009). Information security is a foundational principle in the technology adoption context that focuses on the protection of sensitive information from unauthorized access. Its core focus is ensuring the confidentiality, integrity, and availability of information assets (Guttman and Roback, 1995). As a risk management strategy, information

security confronts various risk dimensions that can arise from technology adoption. During various online transactions, for example, customers are required to provide their personal information, usually recognizing the risk associated with this disclosure (Beldad et al., 2011). Thus, the bare minimum expected from the organizations is that they establish a medium where these transactions are secure and reliable. In a similar vein, providing a secure medium where customers can safely share sensitive information with the service robots is also important for hospitality organizations.

The influence of information security on users' behavioral intentions has been investigated in different contexts. In the e-commerce context, Chang and Chen (2009) defined information security as a component of transaction security, which included the digital payment medium, storage mechanisms, and transmission of all personal information. The findings of the study indicated that, as a component of information security, interface quality was an antecedent to customers' perceptions of security (Chang and Chen, 2009). In the AI context, Jo (2022) evaluated the information security-related factors that affected users' continuance intentions of using artificial intelligence assistants (AIA) (i.e., AI-based software that is capable of performing tasks and services, e.g., Siri, Alexa, Google Now). The findings of the study results indicated two information-related factors, namely trust and privacy concerns on surroundings, to have an impact on users' continuance intentions (Jo, 2022).

As physical entities that engage with their physical environments, including other robots and humans, service robots have the potential to adversely affect customers' use intentions by collecting information through unsecured mediums, such as unencrypted Wi-Fi networks and file transfer protocols (FTPs) that do not use encryption. During these interactions, customers are not expected to follow a particular script since robots can engage in unscripted dialogues while performing the assigned tasks (Paluch et al., 2022). Hiller (2015) emphasized the lack of discussion in the hospitality industry regarding the vulnerability of hospitality information systems (IS) to cyber-attacks. The most common thread in hospitality industry breaches is the hotels' point of sale (POS) systems since not every hotel property is equipped with state-of-the-art security software. Customers' personal information and payment data are often shared in the integrated POS environment and transmitted to the back office first, as opposed to being transmitted to the payment processor directly (Netsurion, 2022).

Another backdoor for hackers to abuse these systems is attacking the Wi-Fi networks. Most hotels these days offer free Wi-Fi access to their customers with network availability in all public spaces, i.e., lobby, food & beverage outlets, banquet spaces, and guest rooms (Shabani and Munir, 2020). Service robots interface with the POS systems and connect to the internet via the same Wi-Fi networks, which may not be as cutting-edge and advanced as the robots themselves,

thus underscoring the crucial role of technology infrastructure in ensuring information security. The cyber-attacks on the hospitality organizations' information systems also demonstrate the significance of cyber security from the perspective of service robot implementation (Makridakis, 2017). In addition to the cyber security protocols, which are a prerequisite for hospitality information systems, including the cloud platforms, POS systems, and Wi-Fi networks, a second layer of cyber security is now necessary for service robots (McCartney and McCartney, 2020). This will not only assure hotel customers that the information they provide to service robots is protected and secured but also promote their willingness to use robots in a hotel setting. Accordingly, the following hypothesis is proposed:

H₂: Hotel customers' perceived information security toward service robots positively influences their intention to use service robots.

2.4. Self-efficacy

Bandura's (1986) social cognitive theory postulates that learning occurs with reciprocal interaction of personal factors (e.g., cognitive

abilities, affective responses), environmental (i.e., extrinsic) influences, and behavior. Self-efficacy, one of the key concepts in social cognitive theory, is defined as "people's judgments about their capabilities to organize and execute courses of action necessary to perform a given task and it is concerned not with the skills one has but with judgments of what one can do with whatever skills one possesses" (Bandura, 1986, p. 391). Similarly, Compeau and Higgins (1995) defined self-efficacy as the individual's belief about his or her ability to capably perform a given or chosen task.

Self-efficacy has been previously proposed as a mitigator of perceived risk and information security concerns as it enhances customers' level of self-confidence (Brody et al., 1988; Kim et al., 2018; Rhee et al., 2009). Kim and Kim (2005) studied self-efficacy in the electronic commerce context and found that self-efficacy had a negative impact on perceived risk. Luo et al. (2010) expounded that self-efficacy functioned as a personal trait factor to nullify customers' perceptions of risk in the acceptance of emerging technologies. In another study, Tamjidyamcholo et al. (2013) investigated information security professionals' behavioral intentions in virtual communities and found a positive relationship between self-efficacy and information security, which transpired through their security knowledge sharing.

A comprehensive assessment of the self-efficacy levels of customers is imperative for hospitality organizations to successfully evaluate their customers' perceptions of risk and information security, and consequently, their intention to use new technologies as seen with the healthcare robot studies (Latikka et al., 2019; Turja et al., 2019). This is also relevant in the context of hotel customers' use intentions of service robots. Given the rise in the deployment levels of artificially intelligent service agents (epitomized by service robots in a physical form) by hospitality organizations (Choi and Wan, 2021), customers' self-efficacy levels could be a determining factor in their risk and information security perceptions of the service robots, which consequently affect their use intentions of these robots.

This study focused on whether the hotel customers believed they were capable of or had the ability and skills necessary to use and interact with service robots. Accordingly, we hypothesize that self-efficacy is negatively associated with perceived risk and positively associated with information security. These relationships may transpire because of customers' confidence in using service robots and belief in their ability to successfully interact with these physical entities without making any mistakes during the provision of their personal information. Based on this theoretical and empirical background, the following hypotheses are proposed:

H₃: Hotel customers' service robot self-efficacy negatively influences their perceived risk toward service robots.

H₄: Hotel customers' service robot self-efficacy positively influences their perceived information security toward service robots.

2.5. Innovativeness

Steenkamp et al. (1999) defined innovativeness as the proclivity to try and purchase new products and brands as opposed to holding on to the same goods and products. In the information technologies context, personal innovativeness was first introduced by Agarwal and Prasad (1998). However, it was previously introduced in the theory of diffusion of innovations as a defining individual trait of early adopters by Rogers (1983, 1995). According to Rogers (1983, 1995), these early adopters also served as critical change agents in propagating the acceptance of new technologies. The theoretical foundation for this proposition was provided by the early works of Bruner and Tajfel (1961) and Jacoby (1971), who were able to establish a relationship between innovativeness and other personality traits.

Makki et al. (2016) advanced the innovativeness literature by empirically examining the relationship between consumers' innovativeness and their intentions to use near-field communication (NFC) technology-based mobile payment systems in the restaurant industry.

The findings of their study indicated that innovativeness was a predictor of consumers' intention to use NFC-based mobile payment systems (Makki et al., 2016). These findings were supported in the robotic context by Nomura et al. (2008) in an experimental design study that involved the interactive humanoid robot 'vie'. Regarding the relationship between consumer innovativeness and perceived risk, Truong (2013) described innovative consumers as risk-takers as they tend to display risk-taking behavior. The present study maintains the same perspective in the robotic context. Innovative hotel customers are expected to look for ways to experiment with service robots despite the risks that might be involved. The same is hypothesized with their perceptions of information security as they provide sensitive information to the robots during their service interactions. In light of this theoretical and empirical support provided by the extant literature for the relationship between innovativeness and perceived risk and information security, the present study proposes the following hypotheses:

H₅: Hotel customers' innovativeness negatively influences their perceived risk toward service robots.

H₆: Hotel customers' innovativeness positively influences their perceived information security toward service robots.

2.6. Facilitating condition

In the context of information technologies, facilitating condition can be defined as the level of credence the users attribute to an infrastructure, which they believe the organization will provide to support the use of a system. The infrastructure, in this context, is provided by the organization to eliminate the barriers to use (Venkatesh et al., 2003). The unified theory of acceptance and use of technology (UTAUT) (Venkatesh et al., 2012) defined facilitation conditions as "consumers' perceptions of the resources and support available to perform a behavior" (p. 161). The positive impact of facilitation conditions on behavioral intention to use technology has been substantiated by the existing literature (Lin et al., 2014; Kim et al., 2020).

In the current study, facilitating condition is conceptualized as the resources necessary to use service robots in a hotel setting. This conceptualization also incorporates the compatibility of the hotels' robotic services with the technologies customers are currently using. A review of the existing literature points to the significance of the facilitating conditions in the robotic context, which has been acknowledged in various contexts. Alaiad and Zhou (2014), for example, explored facilitating conditions' impact on healthcare robot usage intention and found that, as a sociotechnical aspect, facilitating conditions had a positive impact on users' intentions to use robots.

Also in the robotic context, Lu et al. (2019) developed a service robot integration willingness (SRIW) scale to identify the dimensions that characterized consumers' long-term willingness to use service robots. The findings of the study revealed that facilitating conditions significantly affected consumers' willingness to use service robots (Lu et al., 2019). Further, Lu et al. (2019) argued that technologically challenged customers may anticipate a certain level of risk with new and unfamiliar technologies, which may be mitigated by facilitating conditions, such as getting help from other humans. On the other hand, organizations' technological infrastructure and other resources they extend to their customers (e.g., employee assistance) are important for users/customers to feel safe with providing their sensitive information to physical, non-human entities, such as robots, and non-physical, AI-based software. In this regard, it is logical to expect that when hotel customers believe they have the resources necessary to use service robots and can get help from others when they encounter any difficulties when interacting with these robots, they will also assume that they have higher levels of information security. Drawing on this argument, the present study hypothesizes a negative relationship between the facilitating conditions and the perceived risk of using service robots and a positive relationship between facilitating conditions and perceived information security. Against this backdrop, the following hypotheses are proposed:

H₇: Facilitating conditions negatively influence hotel customers' perceived risk toward service robots.

H₈: Facilitating conditions positively influence hotel customers' perceived information security toward service robots.

The theoretical framework below illustrates the proposed hypotheses.

3. Methodology

3.1. Measures

A web-based self-administered online questionnaire was used to collect the data for the study. The questionnaire used in the current study was comprised of three parts. The first part included a definition of robot and an explanation of robotics technologies in the hotel industry with examples. The target population of the study was hotel guests. Therefore, the first part of the questionnaire also included a screening question (i.e., have you stayed in a hotel at least once in the last 12 months). The second part of the questionnaire involved questions related to study constructs (i.e., self-efficacy, innovativeness, facilitating condition, information security, perceived risk dimensions, and intention to use). The last part of the questionnaire included questions related to the participants' demographic characteristics (i.e., gender, age, education, and income).

Previously validated scales were used for all constructs in the current study. Innovativeness was measured with 3 items adapted from Agarwal and Prasad's (1998). Facilitation condition and self-efficacy constructs were measured with 3 items each and they were adapted from Venkatesh et al. (2003) and Compeau and Higgins (1995) respectively. Performance risk was adapted from Stone, Grønhaug (1993) and Sweeney et al. (1999), financial risk was adapted from Featherman and Pavlov (2003) and Stone, Grønhaug (1993), and physical risk and psychological risk scales were adapted from Stone, Grønhaug (1993). All risk scales (dimensions) were measured with 3 items each. A 3-item scale was utilized to measure perceived information security, which was borrowed from Chang and Chen (2009). Finally, intention to use was measured with 3 items adapted from Davis et al. (1992). A seven-point Likert scale ranging from 1 = strongly disagree to 7 = strongly agree was used to measure all study constructs.

In this study, we collected data from 11 different countries representing North and South America, Europe, Asia, and the Middle East. More specifically the data of the current study were collected from the United States, United Kingdom, Turkey, Spain, Romania, Japan, Israel, India, Greece, Canada, and Brazil. The surveys for the United States, United Kingdom, Turkey, Romania, Israel, India, Greece, Canada, and Brazil were prepared in English. The surveys (scales) for Japan and Spain were translated from English to Japanese and Spanish languages by using a back-translation method.

A pilot test was conducted to assess the comprehensiveness and clarity of the questionnaire. For this purpose, we shared the questionnaire with industry experts and faculty members from the related fields and asked them to share their opinions about the survey. One of the common types of feedback they provided was related to the length of the survey. They mentioned that the survey was too long. Another comment was associated with the risk items. The industry experts mentioned that some of the items related to perceived risk constructs were not clear. We shortened the survey and revised the items by making wording modifications based on the pilot study participants' comments and suggestions.

3.2. Data collection and data analyses

The data for Turkey, the United States, Canada, and Spain were collected through a market research company, Qualtrics, and the data for Greece, the United Kingdom, Romania, India, Israel, Brazil, and Japan were collected via local market research companies. The

companies distributed the web-based survey to their randomly selected panel members based on our study's target population (i.e., individuals who stayed in a hotel at least once in the last 12 months). A total of 1545 surveys were collected in October 2020.

Before the data analyses, the data of the study were first inspected for missing values and outliers. After this initial inspection, 23 respondents were removed from the database due to high missing values and outliers, invariance in responding to all the questions, or following a specific pattern of missing values. In the second step, surveys with low missing values (i.e., only a few) were first checked to confirm that the values were missing completely at random and then imputed by the expectation maximization method. After handling the missing data, 1522 surveys were retained for the data analyses (US N = 105, Canada N = 150, Israel N = 238, Brazil N = 127, Spain N = 112, Japan N = 301, Greece N = 128, India N = 52, Turkey N = 107, UK N = 101, Romania N = 101).

For SEM models, sample size is often assessed based on the number of observed variables. For example, a ratio as low as five cases per variable is accepted to be sufficient when latent variables have multiple indicators (Bentler & Chou, 1987). A generally accepted rule of thumb, however, is 10 cases per observed variable in setting a lower bound of an adequate sample size (Hair et al., 2010). Hence the sample size of this study was meaningful for analyzing the measurement and the structural models.

Histograms and box plots were visually inspected to evaluate the normality of the dataset. The results indicated that the variables used in the current study demonstrated normal distribution. In addition, skewness and kurtosis values were also examined to confirm whether the study data were normally distributed. The results demonstrated that skewness and kurtosis values were all in the recommended range of -2 and $+2$ (George & Mallery, 2010). Furthermore, multicollinearity was assessed by calculating tolerance values. The results indicated that the tolerance values of each construct were higher than 0.2 (Hair et al., 2010) demonstrating no multicollinearity issues in the present study.

In addition, common method bias was tested by conducting Harman's one-factor statistical test. According to Harman's one-factor test, an exploratory factor analysis on SPSS was performed by constraining the number of factors extracted to one factor with no rotation. The results revealed that common method bias was not an issue in the current study since less than 50% of the variance (40.2%) was explained by a single factor (Podsakoff et al., 2003).

The demographic characteristics of the respondents are presented in Table 1. The study data revealed that 43% of the respondents were male and around 30% of the participants were between the ages of 18 and 55. Regarding the education level, around 30% of the respondents had a bachelor's degree. Finally, 36% of the participants had above-average income.

In this study, we adopted a two-step approach was followed for the data analyses. The first step involved conducting a confirmatory factor analysis (CFA) to evaluate the model fit for the first-order measurement model and to assess the validity of the study scales. In the current study, perceived risk was conceptualized as a reflective second-order construct, which is comprised of physical risk, financial risk, psychological risk, and performance risk. Therefore, the first step of the data analysis also involved performing an additional CFA to assess the model fit for the second-order measurement model with perceived risk as a second-order construct. In the second step, a covariance-based structural equation modeling (CB-SEM) was utilized to test the study hypotheses (Hair et al., 2017). By adopting a confirmatory approach and utilizing previously validated scales, the main purpose of our study was to test and confirm a research model that was based on existing theories. In other words, we did not develop a theory or heavily revise an existing one to structure our research model. Therefore, we employed a CB-SEM as an appropriate data analysis technique to test the study hypotheses (Dash et al., 2021).

Table 1
Demographic Profile of the Respondents.

Demographics	N	%
Gender	660	43.4
Male	703	46.2
Female	22	1.4
Prefer not to say	137	9.0
Missing	1522	100.0
Total		
Age	263	17.3
18–25	380	25.0
26–35	294	19.3
36–45	240	15.8
46–55	139	9.1
56–65	65	4.3
66 or older	7	0.5
Prefer not to say	134	8.8
Missing	1522	100.0
Total		
Education Level	125	8.2
High school	76	5.0
Associate degree (2 year)	289	19.0
Some college	425	27.9
Bachelor's degree (4 year)	319	21.0
Master's degree	123	8.1
Doctorate degree	28	1.8
Prefer not to say	137	9.0
Missing	1522	100.0
Total		
Household Income	549	36.1
Above the average	369	24.2
At average	377	24.8
Below the average	87	5.7
Prefer not to say	140	9.2
Missing	1522	100.0
Total		

4. Results

The results of the CFA revealed an acceptable model fit for the measurement model with the ratio of chi-square (χ^2) value to degree of freedom value of 2.968. In addition, other model fit indices including GFI (0.93), NFI (0.96), CFI (0.94), and RMSEA (0.043) also demonstrated an acceptable model fit (Hair et al., 1998).

Cronbach's alpha and composite reliability (CR) scores were calculated to check the study scales' reliability. The α coefficients for the study scales ranged between 0.80 and 0.92. The results indicated that the CR values were all above 0.7, which illustrated that all of the study scales were reliable (Nunnally, 1970).

Furthermore, to determine the convergent and discriminant validities, average variance extracted (AVE) values, and the comparison of squared root of AVEs with correlations between constructs were utilized respectively (Fornell and Larcker, 1981). The results indicated that the AVE values of the constructs were between 0.51 and 0.76, which were greater than the recommended cutoff value of 0.50 and the correlations between each construct were less than the squared root of AVE values, demonstrating that the convergent and discriminant validities were established (Tables 2 and 3) (Hair et al., 2010). In addition, the heterotrait–monotrait (HTMT) ratio of correlations was also calculated to assess the discriminant validity of the constructs. The results indicated that all of the HTMT values were less than 0.85 confirming that discriminant validity was not an issue in this study.

As mentioned earlier, perceived risk was conceptualized as a second-order construct which is comprised of physical risk, financial risk, psychological risk, and performance risk. The results of the second-order CFA demonstrated an acceptable model fit with $\chi^2/df = 2.899$; GFI = 0.94; CFI = 0.96, NFI = 0.91, and RMSEA = 0.49. The CR value for perceived risk as a second-order construct was 0.89 demonstrating that the scale was reliable. In addition, considering the AVE value and the inter-construct correlations (i.e., correlations among perceived risk

Table 2
Confirmatory Factor Analysis Results for the First Order Measurement Model.

Constructs	Loadings	CR	AVE
Innovativeness		0.860	0.672
1. If I heard about a new information technology, I would look for ways to experiment with it.	0.846		
2. Among my peers, I am usually the first to explore new information technologies.	0.807		
3. I like to experiment with new information technologies.	0.806		
Facilitating Conditions		0.774	0.636
1. I have the recourses necessary to use robots in hotels.	0.745		
2. robots in hotels are compatible with other technologies I use.	0.734		
3. I can get help from other people when I have difficulties using robots in hotels.	0.730		
Self-Efficacy		0.859	0.675
1. I would use robots in hotels if only there is a built-in help facility for assistance.	0.903		
2. I would use robots in hotels if someone else had helped me to get started.	0.895		
3. I would use robots in hotels if someone showed how to do it first.	0.739		
Performance Risk		0.759	0.643
1. Being served by robots in hotels would not turn out as expected.	0.768		
2. There is a high chance that there would be something wrong with being served by robots in hotels.	0.763		
3. Overall, being served by robots in hotels is risky in terms of how it would perform.	0.701		
Financial Risk		0.814	0.594
1. If I am served by robots in hotels, I would be concerned that it could lead to a financial loss for me.	0.772		
2. Being served by robots in hotels would be a bad way to spend my money.	0.813		
3. If I am served by robots in hotels, I would be concerned that I really would not get my money's worth from this service.	0.724		
Physical Risk		0.802	0.620
1. Being served by robots in hotels could lead to some uncomfortable physical side-effects.	0.771		
2. Being served by robots in hotels would make me concerned about potential physical risks.	0.763		
3. I would feel physically unsafe if I am served by robots in hotels.	0.740		
Psychological Risk		0.909	0.769
1. The thought of being served by robots in hotels makes me feel psychologically uncomfortable.	0.881		
2. The thought of being served by robots in hotels gives me a feeling of unwanted anxiety.	0.812		
3. The thought of being served by robots in hotels causes me to experience unnecessary tension.	0.901		
Information Security		0.902	0.755
1. robots provide a secure medium to send sensitive information.	0.903		
2. I would feel totally safe to provide sensitive information about myself over robots.	0.910		
3. Overall, robots are a safe place to transmit sensitive information.	0.787		
Intention to Use		0.937	0.831
1. Given a chance, I intend to use a robot application in a hotel.	0.906		
2. Given a chance, I predict that I should use a robot application in a hotel.	0.912		
3. Given a chance, I plan to use a robot application in a hotel.	0.918		

dimensions), it is determined that the convergent and discriminant validities were confirmed for the second-order model (Fornell & Larcker, 1981) (Tables 4 and 5). {{Fig. 1}}.

After confirming the model fit for the first-order and the second-order measurement models, and the validity and reliability of the scales, we proceeded with testing the structural model. The results indicated an acceptable model fit for the structural model (i.e., $\chi^2/df = 3.114$, GFI = 0.95, CFI = 0.93, NFI = 0.95, and RMSEA = 0.048). The hypothesis testing results indicated that all of the hypotheses except H5 and H7 were supported. More specifically, the study results demonstrated that perceived overall risk had a negative and information security had a positive impact on intention to use service robots (H1 and H2 were supported). In addition, the study results indicated that self-efficacy negatively influenced perceived overall risk, and positively influenced perceived information security (H3 and H4 were supported). Furthermore, the study results revealed that innovativeness and facilitating condition were positively associated with information security (H6 and H8 were supported) (Table 6, Fig. 2).

5. Conclusion and discussion

This study provides a comprehensive analysis of the factors that influence customers' intentions to use robotic services in the hotel industry. Specifically, the study proposes a conceptual model that highlights the role of perceived risk and information security in shaping customers' intentions to use robots. To enrich the findings, the authors also tested the effects of self-efficacy, innovativeness, and facilitating conditions on perceived risk and information security. These variables are important because they have been shown to influence customers' perceptions of the benefits and risks of using new technologies (e.g., Makki et al., 2016; Alalwan et al., 2016; Keith et al., 2015).

The model testing results indicate that self-efficacy had significant impacts on both perceived risk and information security, which collectively affected hotel customers' intentions to use robotic services. This finding suggests that customers who have high levels of self-efficacy may be more likely to perceive robotic services as secure and less risky, which in turn increases their intentions to use them. Furthermore, the study's findings are consistent with previous literature such as Chang (2021) on the retail shopping experience and Rhee et al. (2009) from the perspective of social cognitive theory. This consistency adds to the validity and generalizability of the study's findings.

Moreover, the variable of innovativeness was found to significantly influence information security, while not showing a statistically significant effect on perceived risk. The effect of innovativeness on information security receives supporting evidence from Jaafar and Ajis's (2013) work in organizational climate research. The possible reasons for the insignificant effects of innovativeness on perceived risk could include the situations where innovative people tend to have adequate knowledge of new things (Rogers, 1983, 1995), which leads them to be early adopters of new products and services, such as robotic services. In these situations, innovativeness seems irrelevant to risk perceptions.

Facilitating condition was found to have a significant effect on information security rather than perceived risk. Facilitating conditions denote the infrastructure (Venkatesh et al., 2003) that ensures the customers that their private information is securely protected when they give their personal information to physical entities, such as service robots, or non-physical AI-based software, such as Siri, Alexa, and Google Now. However, risk perceptions emphasize the uncertainty the customers have towards the robotic servers, which seems not significantly related to whether the infrastructure protects their information privacy or not. This could possibly be one of the reasons why facilitating conditions didn't affect perceived risk in our study.

Table 3
Discriminant Validity for the First Order Measurement Model.

Constructs	1	2	3	4	5	6	7	8	9
1. Inf. Security	0.869								
2. Physical Risk	-0.029	0.788							
3. Financial Risk	-0.005	0.629	0.771						
4. Performance Risk	-0.025	0.577	0.604	0.802					
5. Psychological Risk	-0.080	0.808	0.525	0.506	0.877				
6. Self-Efficacy	0.371	-0.139	-0.069	-0.263	-0.072	0.822			
7. Innovativeness	0.510	-0.001	-0.070	-0.051	-0.132	0.395	0.820		
8. Facilitating Cond.	0.371	-0.228	-0.083	-0.175	-0.120	0.349	0.536	0.798	
9. Intention to Use	0.515	-0.155	-0.194	-0.027	-0.272	0.521	0.676	0.538	0.912

Note: Off-diagonal elements: Inter-construct correlations. Diagonal elements (bold): Squared root of AVEs

Table 4
Second Order Measurement Model Results.

Constructs	Loadings	CR	AVE
Perceived Overall Risk		0.89	0.69
1. Physical Risk	0.949		
2. Financial Risk	0.870		
3. Performance Risk	0.731		
4. Psychological Risk	0.846		

Table 5
Discriminant Validity for the Second Order Measurement Model.

	1	2	3	4	5	6
1. Self- Efficacy	0.821					
2. Innovativeness	0.394	0.820				
3. Facilitating Cond.	0.351	0.537	0.715			
4. Inf. Security	0.371	0.510	0.372	0.869		
5. Intention to use	0.521	0.675	0.540	0.515	0.912	
6. Overall Risk	-0.131	-0.058	-0.178	-0.043	-0.215	0.832

Overall, the proposed conceptual model and the findings of this study provide valuable insights into the factors that influence customers' intentions to use robotic services in the hotel industry. These insights can be used by hotel managers to design and implement robotic services that meet the needs and expectations of their customers. Moreover, the study's contributions to the existing literature on customers' perceptions of risk and the security of new technologies have important implications for future research in this area. The current study's contributions to academia and practice are elaborated in detail as follows.

5.1. Theoretical implications

First, this study proposed and validated the four sub-dimensions of perceived risk in the context of robotic services in hotels. These four sub-dimensions were physical risk, financial risk, performance risk, and psychological risk. This is a significant contribution as prior literature in the domain of perceived risks has revealed various explaining dimensions across technology-facilitated services or products. For example, Hwang and Choe (2019) identified five facets underlining perceived risks, including financial, time, privacy, performance, and psychological risks, when studying drone food delivery services. Martins et al. (2014) found that social risks could be the new addition to the perceived risk dimensions, compared to the traditional components of financial, performance, privacy, and time risks in the context of service automation systems in the banking services. In the field of mobile app adoption, Park and Tusssyadiah (2017) confirmed device risks as the most salient factor forming the overall risks of mobile booking intentions. The findings of this study contribute to the current understanding of perceived risks, especially the sub-dimensions of the construct, under the research background of service robots in the hotel industry.

Second, the study examines the outcome variable of perceived risk and information security using the structural equation modeling technique. The results show that both perceived risk and information security have significant effects on hotel customers' intentions to use service robots. This provides empirical evidence and extends the literature (Qian et al., 2012; Im et al., 2008) by identifying the significant explaining powers of perceived risk and information security in customers' intentions to use service robots in a hotel setting. The results also contribute to the development of theories of perceived risk and information security by identifying their significant roles in shaping customers' intentions to use service robots in a hotel setting.

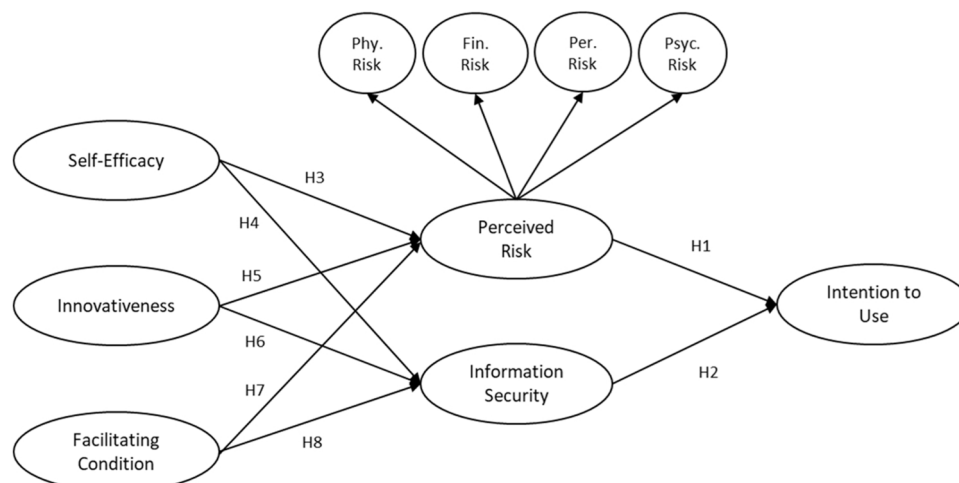


Fig. 1. Research Model.

Table 6
Hypotheses Test Results.

Paths	Standardized Path Coefficients	Hypothesis Supported yes/no
H1: Perceived Risk → Intention to Use	-0.21 **	Yes
H2: Information Security → Intention to Use	0.55 **	Yes
H3: Self-Efficacy → Perceived Risk	-0.12 *	No
H4: Self-Efficacy → Information Security	0.20 **	No
H5: Innovativeness → Perceived Risk	0.07 n.s.	No
H6: Innovativeness → Information Security	0.45 **	Yes
H7: Facilitating Condition → Perceived Risk	0.07 n.s.	No
H8: Facilitating Condition → Information Security	0.15 *	Yes

Note: * $p < 0.05$; ** $p < 0.01$; n.s. non-significant

Third, this study is among the first to simultaneously examine customers' personal traits and technology infrastructure conditions on customers' perceived risks and information security for robotic services in a hotel setting. Although prior studies primarily focused on service robot attributes, for instance, anthropomorphism (Lu et al., 2021), warmth (Zhu & Chang, 2020), humor (Zhang et al., 2021), and interactivity (Shin and Jeong, 2020), this study is a pioneering attempt to unveil the influence of customers' personal characteristics (innovativeness, and self-efficacy) and facilitating conditions provided by the new technology-based services on customers' perceived risks and information security when adopting robotic services. The study findings validate the relevance of self-efficacy to customers' perceived risks and information security, whereas innovativeness and facilitating conditions only significantly relate to information security rather than perceived risks. This novel finding adds to the knowledge based on individuals' characteristics and human-t interactions.

5.2. Managerial implications

First, hotel managers are advised to consider the four major facets (physical risks, financial risks, performance risks, and psychological risks) of risks perceived by the customers when designing and introducing service robots in their hotels. When hotel managers integrate service robots, they must recognize that customer perceptions are deeply influenced by habits, biases, and heuristics. For example, an unfamiliar robot design might evoke psychological risks, akin to unsettling sci-fi movies. Thus, robots should have intuitive, friendly designs, and guests can be educated about them through compelling storytelling or

interactive sessions. Leveraging innate human preferences, like a robot humming a familiar tune, can enhance acceptance. Collaborations with behavioral scientists can further optimize robot interactions to resonate with guests, and feedback mechanisms or pilot tests can offer invaluable insights. With the significant investments involved in deploying service robots, aligning them with deep-rooted human behavioral patterns is as essential as technological efficiency.

Second, while the data shows that perceived risks and information security are pivotal for customers adopting robot services, we must also consider the power of human habits and biases. Habitual practices can make customers hesitant about robot services. For instance, when robots perform functions that were traditionally done by humans like check-in/out or concierge services, customers, due to their habitual biases, might view this unfamiliar method with skepticism, particularly when required to share personal information. Therefore, to alleviate concerns rooted in biases and heuristics, hoteliers should not only upgrade their security infrastructure but also employ clear, relatable communication about data privacy, perhaps using relatable anecdotes or stories to make it more comprehensible and reassuring.

Third, targeting individuals with high self-efficacy and innovativeness might work wonders, given the influence of positive biases toward novel technology. By presenting service robots as advanced yet user-friendly, hotels can leverage the cognitive shortcut of "novelty as progress," encouraging patrons to feel more secure and less risk-averse. Moreover, a seamless technology infrastructure plays into the heuristic of ease and familiarity, further reducing perceived risks. For example, The Ritz-Carlton hotel could introduce an AI-powered concierge named "RitzBot" that scans guests' public social media profiles (with prior permission) to curate a list of activities or restaurant suggestions tailored to their interests. For an individual with high self-efficacy and a penchant for technology, this personalized touch, fueled by advanced technology, can enhance their stay, making them feel catered to and understood.

Fourth, while emphasizing stringent information security measures is paramount, hotel managers should also understand that customers' trust is influenced by the availability heuristic. Offering transparent, easy-to-grasp insights about data protection and the tangible benefits of robotic services will help patrons remember and value these positive aspects, instead of potential risks. For example, if hotels integrated a feature in their robot-assisted room service where, before payment details are requested, a brief animated video plays, showing how their data is encrypted and protected, it would allay fears. This immediate, visual reassurance leans into the availability heuristic, placing positive, trust-inducing information directly in the moment of potential doubt.

Fifth, self-efficacy does play a significant role, but it's vital to

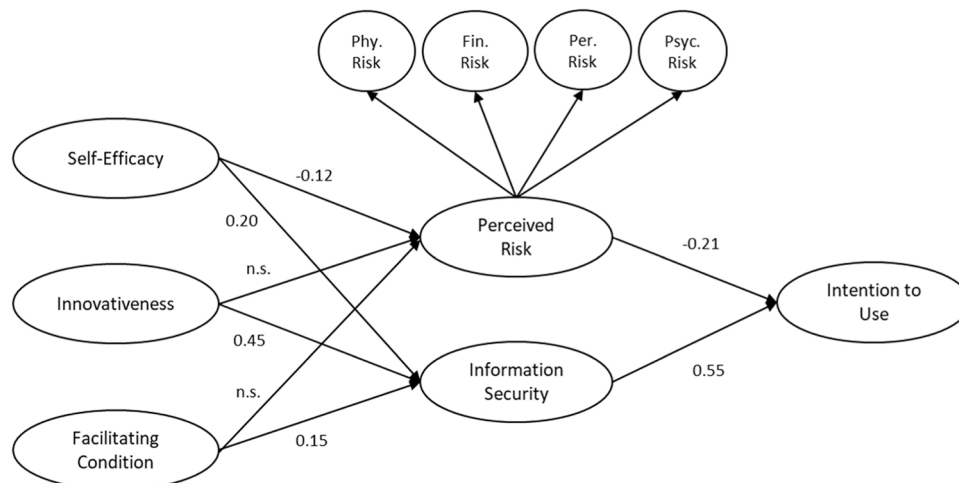


Fig. 2. Results for the Structural Model Analysis.

remember that confidence often stems from familiarity. Training programs for employees should, therefore, not just impart skills but also habitualize them to robotic services. Continuous customer feedback loops can further cater to the recency bias, ensuring the latest concerns are always addressed, thereby maintaining trust and enhancing the overall customer experience with robotic services in hotels. Consider a hotel introducing a robotic bellhop service. Before fully rolling it out, they could host weekly "Introduction Sessions" for their staff. Here, employees would not only learn how to operate the robot but would also role-play as guests, experiencing firsthand the robot's service offerings. Over time, this consistent interaction transforms the unfamiliar into the familiar, making staff more confident in promoting and explaining these services to guests. Additionally, setting up kiosks in the lobby where guests can provide instant feedback on their robotic service experience, or even suggest improvements, ensures the service stays relevant and customer-centric. This real-time feedback mechanism addresses any concerns immediately, making guests feel heard and valued.

6. Limitations and future study

This study has several limitations. Therefore, the study findings should be interpreted carefully. For instance, a perception-based approach was adopted in the present study and hotel customers' real service robot usage behavior was not considered. In addition, this study mainly focused on hotel customers' service robot usage intentions by examining the impacts of a limited number of antecedents. In this regard, future research, which should investigate the influences of other possible antecedents (e.g., service quality, personalization, trust) on hotel customers' actual behavior measures may provide more valid results for hotel operators and service robot vendors.

Even though the data of the study were collected from 11 different countries, the differences in hotel customers' service robot perceptions based on their cultural backgrounds and demographic characteristics were not examined. In addition, considering the increased adoption of service robots by hotel operators around the world, there may be a significant number of hotel guests who may have already experienced service robots in hotels or other settings such as restaurants, cruise ships, or hospitals. However, this study did not examine the differences in hotel guests' service robot acceptance based on their prior experience. Therefore, future studies investigating whether hotel guests' service robot acceptance differs based on their cultural backgrounds, demographic backgrounds, and past experiences may offer useful and valuable insights.

The study focused on gauging intentions rather than observing actual behaviors. As Juvan and Dolnicar (2014) have highlighted, there can often be discrepancies between what individuals intend to do and their eventual actions. It would be beneficial for future research to conduct field studies, assessing the real-world behaviors of participants when interacting with robotic services.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data Availability

The authors are unable or have chosen not to specify which data has been used.

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